

Digital Signal Analysis and Processing Exam Notes

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

3. Analysis of LTI Systems in the Frequency Domain

6 hrs · 56 questions across the 45 papers · worth about 8.4 of 80 marks · 380 marks, 10.4% of the archive · 2 syllabus sub-topics · 6th of the 8 chapters by weight: only Ch 2 and the EX 753 extras are lighter.

Read this first:

- 35 of the 56 questions are one question, asked again and again with the numbers changed: *plot the poles and zeros in the z -plane, then sketch the magnitude response, not to scale* TOP 35/45. It is worth 6 to 10 marks and it is the single most valuable thing in this chapter.
- It arrives in two disguises. Either a **difference equation** is given and you must find $H(z)$ first, or the **poles and zeros are handed to you** directly, sometimes in polar form. The second half of the work is identical.
- The remaining 21 questions are short bookwork worth 2 to 4 marks each: causality and stability from the impulse response and the ROC HOT 7/45, the LCCDE and its system function HOT 6/45, FIR against IIR PIN 3/45, linear phase PIN 3/45, zero-input and zero-state response PIN 2/45.
- **Sketching does not mean guessing.** Three numbers, $|H|$ at $\omega = 0, \pi/2$ and π , plus the pole and zero angles, fix the shape completely. §3.3 gives the recipe.
- The cheapest check in the chapter: $H(e^{j0}) = H(1)$ is the sum of all coefficients and $H(e^{j\pi}) = H(-1)$ is their alternating sum. Both are one line of arithmetic, and both are marks.

3.1 Frequency response of an LTI system

Derive the expression for the frequency response of an LTI system. Show that a complex exponential is an eigenfunction of an LTI system 6

PIN 2/45 69 Bh, 67 Mng · the opening paragraph of most answers in this chapter

The eigenfunction property

Feed a complex exponential into an LTI system with impulse response $h[n]$ and convolve:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] e^{j\omega(n-k)} = e^{j\omega n} \underbrace{\sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k}}_{H(e^{j\omega})}$$

- The input comes out unchanged in *shape*, scaled by a complex number. So $e^{j\omega n}$ is an **eigenfunction** of every LTI system and $H(e^{j\omega})$ is the **eigenvalue**.
- $H(e^{j\omega})$ is the **frequency response**: the DTFT of $h[n]$, and equally $H(z)$ evaluated on the unit circle.

$$H(e^{j\omega}) = H(z) \Big|_{z=e^{j\omega}}$$

- **This substitution is legal only if the ROC contains the unit circle.** If it does not, the system has no frequency response at all, however neatly $H(z)$ factors. See §3.4.

Magnitude, phase and group delay

- Writing $H(e^{j\omega}) = |H(e^{j\omega})| e^{j \arg H(e^{j\omega})}$, the three standard plots are the **magnitude response** $|H|$, the **phase response** $\arg H$, and the **group delay**

Response to a real sinusoid

For $x[n] = A \cos(\omega_0 n + \phi)$ and a real $h[n]$, superposing the two exponentials gives

$$y[n] = A |H(e^{j\omega_0})| \cos(\omega_0 n + \phi + \angle H(e^{j\omega_0}))$$

- The system can only do two things to a sinusoid: **scale it** by $|H|$ and **shift it** by $\angle H$. It can never change its frequency.
- That is the whole reason the magnitude and phase responses are worth plotting.

Three properties that halve the work

- $H(e^{j\omega})$ is **periodic in ω with period 2π** , because $e^{j\omega n}$ is.
- For real $h[n]$, $|H|$ is **even** and $\angle H$ is **odd** in ω .
- So **everything is known from $0 \leq \omega \leq \pi$** , and that is the only range any paper asks you to draw.

$$\text{grd } H(e^{j\omega}) = -\frac{d}{d\omega} \arg H(e^{j\omega})$$

- Group delay is the delay the system applies to the envelope of a narrow band of frequencies. **Constant group delay means linear phase means no phase distortion**: see §3.6.
- Log magnitude is quoted in dB as $20 \log_{10} |H(e^{j\omega})|$.

3.2 The LCCDE and its system function

State the linear constant coefficient difference equation and the corresponding system function. Why do we need a difference equation? **3 2**

HOT 6/45 73 Shr **3**, 71 Shr **3+2**, 69 Ch **2**, 69 Bh **2**, 73 Ch **2** · pure bookwork, **same answer every year**

The equation

An N th-order linear constant-coefficient difference equation relates the output samples to the input samples with coefficients that do not change with n :

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

- **Linear**: no products or powers of x or y , no $\sin y[n]$.
- **Constant coefficient**: a_k and b_k are numbers, not functions of n . This is what makes the system time-invariant.
- **Order N** : how many past outputs are fed back.
- Normalising $a_0 = 1$ gives the recursive computing form, which is how it is actually run on hardware:

$$y[n] = \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k]$$

The system function

Z-transform both sides, using linearity and the time-shift property $y[n-k] \leftrightarrow z^{-k}Y(z)$:

$$Y(z) \sum_{k=0}^N a_k z^{-k} = X(z) \sum_{k=0}^M b_k z^{-k}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} = \frac{B(z)}{A(z)}$$

- **An LCCDE always gives a rational $H(z)$** , and every rational $H(z)$ comes from some LCCDE. The two descriptions are the same object.
- **Zeros** come from the x side (feedforward), **poles** from the y side (feedback). Read them straight off the two coefficient lists.
- Multiply top and bottom by z^N to clear the negative powers before factoring.

Why we need a difference equation (a 2 mark question, asked five times)

- It is a **complete description** of an LTI system in the time domain, finite and exact, whereas $h[n]$ may be an infinitely long list.
- It is **recursive**: each output uses a handful of stored past values, so an IIR response of infinite length costs finite memory and finite arithmetic.
- It maps **directly onto hardware**: every term is one delay, one multiplier and one adder, which is exactly what a realisation structure draws (Ch 4).
- It gives $H(z)$ **by inspection**, so the frequency response, the poles, the zeros and the stability all follow from the same few coefficients.
- It carries **initial conditions**, which is what separates the zero-input from the zero-state response.

A degree count worth remembering: if $M < N$, then $B(z)$ written in positive powers is short by $N - M$, so $H(z)$ **has $N - M$ zeros at the origin**. They are real zeros and they belong on the plot, but being at $z = 0$ they are equidistant from every point of the unit circle and so **contribute nothing to the magnitude shape**.

3.3 Pole-zero plot to magnitude response

Plot the poles and zeros in the z -plane and sketch the magnitude response (not to scale) of the system described by the given difference equation **3+7 2+8 6**

TOP 35/45 80 Bh, 81 Ba, 76 Ash, 79 Bh, 79 Ba, 78 Bh, 81 Bh, 80 Ch, 72 Ash, 74 Ash, 72 Ka, 72 Ch, 82 Ba, 70 Ch, 71 Bh, 73 Ma, 69 Ch, 76 Bh, 75 Ch, 69 Bh, 67 Mng, 68 Bh, 70 Asa, 70 Ma, 80 Ba, 82 Bh, 76 Ch, 75 Ash, 81 Ch, 73 Bh, 74 Ch, 75 Bh, 73 Ch, 71 Ch · **worked in full in the numerical companion, §1 and §2**

The geometric idea, which is the whole method

Factor $H(z)$ into its zeros z_k and poles p_k :

$$H(z) = K \frac{\prod_{k=1}^M (z - z_k)}{\prod_{k=1}^N (z - p_k)}$$

Now put $z = e^{j\omega}$, a point walking round the unit circle. Taking magnitudes turns the product into distances:

$$|H(e^{j\omega})| = |K| \frac{\prod_k |e^{j\omega} - z_k|}{\prod_k |e^{j\omega} - p_k|}$$

- Every factor is the **length of the vector** drawn from a zero or a pole to the point $e^{j\omega}$.
- **Zero distances multiply on top, pole distances divide underneath.**
- So as ω sweeps 0 to π : passing **near a pole** makes a denominator distance small, and $|H|$ **peaks**; passing **near a zero** makes a numerator distance small, and $|H|$ **dips**.
- A zero exactly on the unit circle gives a distance of zero, so $|H|$ is **exactly zero** there: a perfect null.
- The closer a pole sits to the circle, the taller and narrower its peak.

The algebraic form, for when a paper wants the expression and not only the picture

Each first-order factor, written with the pole or zero at radius r and angle θ , contributes

$$|1 - re^{j\theta}e^{-j\omega}|^2 = 1 + r^2 - 2r \cos(\omega - \theta)$$

- In dB: $20 \log_{10} |1 - re^{j\theta}e^{-j\omega}| = 10 \log_{10} (1 + r^2 - 2r \cos(\omega - \theta))$, **added** for a zero factor and **subtracted** for a pole factor.
- Phase: $\arg(1 - re^{j\theta}e^{-j\omega}) = \arctan \frac{r \sin(\omega - \theta)}{1 - r \cos(\omega - \theta)}$, again + for a zero and – for a pole.
- Group delay of the same factor: $\frac{r^2 - r \cos(\omega - \theta)}{1 + r^2 - 2r \cos(\omega - \theta)}$.
- **The minimum of $1 + r^2 - 2r \cos(\omega - \theta)$ is at $\omega = \theta$** , where it equals $(1 - r)^2$. That is why a pole at angle θ puts the peak of $|H|$ at $\omega = \theta$, and why $r \rightarrow 1$ makes it sharp.

Poles and zeros given in polar form

Several papers, and the local worked example the family is copied from, specify a radius and an angle instead of a complex number. Convert first:

$$z = re^{j\theta} = r \cos \theta + jr \sin \theta$$

For example radius 0.9 at angle 1.0376 rad gives $0.45 + j0.77$, and radius 0.892 at 2.5158 rad gives $-0.728 + j0.522$. **Those are the exact numbers the papers reuse**, so recognising them saves a conversion.

The exam recipe

1. From the LCCDE write $H(z) = B(z)/A(z)$ in z^{-1} , then multiply top and bottom by z^N so both are polynomials in z .
2. **Factor.** Roots of the numerator are zeros, roots of the denominator are poles. Use the quadratic formula: these are almost always second order and the roots almost always come out complex conjugate.
3. **Plot** the unit circle, mark each zero $\circ$ and each pole $\times$, and do not forget the zeros at the origin.
4. **Evaluate three points**, which is all a not-to-scale sketch needs:

$$|H(e^{j0})| = \left| \frac{B(1)}{A(1)} \right|, \quad |H(e^{j\pi/2})| = \left| \frac{B(j)}{A(j)} \right|, \quad |H(e^{j\pi})| = \left| \frac{B(-1)}{A(-1)} \right|$$

5. **Locate the peak** at the angle of the pole, and the dip at the angle of the zero. Join the points smoothly.
6. State the character in words: low-pass, high-pass, band-pass or notch.

$B(1)/A(1)$ is just the sum of the coefficients, and $B(-1)/A(-1)$ the alternating sum. **Two lines of arithmetic for two anchor points of the sketch.**

3.4 Causality and stability, from $h[n]$ and from the ROC

Describe the stability and causality characteristics of an LTI system in terms of its impulse response and the ROC of its transfer function, with suitable examples 4+3 4 2 3

HOT 7/45 76 Ash 4+3, 72 Ch 4, 81 Ch 2, 80 Ch 2, 73 Ma 2, 73 Bh 3, 77 Ch 3 · learn the four-line table and the one-line rule

In terms of the impulse response		In terms of the ROC	
Property	Condition on $h[n]$	Property	Condition on the ROC
Causal	$h[n] = 0$ for all $n < 0$	Causal	exterior of the outermost pole, and includes $z = \infty$
Stable (BIBO)	$\sum_{n=-\infty}^{\infty} h[n] < \infty$	Stable	the ROC contains the unit circle
		Both	all poles strictly inside $ z = 1$
– Causality says the output cannot anticipate the input: it is a statement about <i>when</i> $h[n]$ is allowed to be non-zero. – Stability says a bounded input always gives a bounded output, and the necessary and sufficient test is that $h[n]$ be absolutely summable. – The two are independent. A system can be either, both or neither.		– The last line is the one to quote: a causal system is stable if and only if every pole has $p < 1$. – A pole exactly <i>on</i> the circle is marginally stable, which is not BIBO stable: the frequency response blows up there. – Stability is also why the frequency response may not exist. If the ROC misses the unit circle, $H(e^{j\omega})$ is undefined, so no magnitude response can be drawn at all.	

One $H(z)$, three systems (the standard worked example, and a favourite trap)

Take $H(z)$ with poles at $z = \frac{1}{2}$ and $z = 3$. The algebra is identical in all three cases; only the ROC differs, and the ROC is a separate piece of information the question must supply.

ROC	$h[n]$	Causal?	Stable?
$\frac{1}{2} < z < 3$	$(\frac{1}{2})^n u[n] - 2(3)^n u[-n - 1]$	no	yes
$ z > 3$	$(\frac{1}{2})^n u[n] + 2(3)^n u[n]$	yes	no
$ z < \frac{1}{2}$	$(\frac{1}{2})^n u[-n - 1] + 2(3)^n u[-n - 1]$	no	no

The middle row is the point: **a causal system with a pole outside the unit circle is unstable**, no matter how ordinary $H(z)$ looks. The top row is the only stable reading, and it is not causal.

3.5 FIR against IIR

Differentiate between an FIR system and an IIR system 4

PIN 3/45 80 Ba 4, 76 Bh 4, 67 Mng · a table answer, four or five rows is enough

	FIR	IIR
Impulse response	finite length, $h[n] = 0$ outside $0 \leq n \leq M - 1$	infinite length, decays but never ends
Difference equation	non-recursive, output from inputs only	recursive, past outputs fed back
$H(z)$	all zeros, poles only at the origin	has poles away from the origin
Stability	always stable , a finite sum is always bounded	stable only if every pole is inside $ z = 1$
Linear phase	easy , exact, by making $h[n]$ symmetric	not achievable exactly
Order needed	high, for a sharp response	low, for the same sharpness
Memory, arithmetic	more	less

- The two rows examiners look for are **FIR is always stable** and **FIR can have exactly linear phase**. Those are the reasons to pay for its higher order.
- An FIR system's $H(z) = b_0 + b_1 z^{-1} + \dots + b_{M-1} z^{-(M-1)}$ written over z^{M-1} has all its poles at $z = 0$, which is inside the unit circle: the stability claim needs no further argument.

3.6 Linear phase

What is the condition satisfied by a linear phase FIR filter? Derive the frequency response of a symmetric linear phase filter of length M , M odd 2+4 6

PIN 3/45 70 Asa 2+4, 67 Mng 6, 69 Bh 6 · the only chapter source is **FIR-design material**: this topic is taught again in Ch 5

What linear phase means, and why it is wanted

- Phase is **linear** when $\arg H(e^{j\omega}) = -\alpha\omega$, a straight line through the origin.
- Then the group delay $-\frac{d}{d\omega} \arg H = \alpha$ is **constant**: every frequency is held up by the same α samples.
- A common delay shifts the signal but does not deform it. **Linear phase means no phase distortion**, which matters wherever shape carries the information: speech, ECG, data pulses, images.
- If the phase were curved, different frequency components would arrive with different delays and a pulse would spread out, even with a perfectly flat $|H|$.
- **Generalized** linear phase allows a constant offset, $\arg H = \beta - \alpha\omega$. The group delay is still constant, so it still does not distort; $\beta = \pm\pi/2$ is what an antisymmetric $h[n]$ produces.

The condition

An FIR filter of length M has linear phase if its impulse response is symmetric or antisymmetric about its midpoint:

$$h[n] = \pm h[M-1-n], \quad 0 \leq n \leq M-1$$

- + gives **symmetric** $h[n]$ and phase $-\alpha\omega$ with $\alpha = \frac{M-1}{2}$.
- – gives **antisymmetric** $h[n]$ and phase $\frac{\pi}{2} - \alpha\omega$, with the same α .
- $\alpha = (M-1)/2$ is **the centre of the sequence**, and it is the delay in samples. For odd M it is an integer; for even M it is a half-integer, so the filter delays by half a sample.
- In the z -plane the condition forces zeros to occur in conjugate reciprocal groups: if z_0 is a zero, so are z_0^* , $1/z_0$ and $1/z_0^*$.

Derivation for symmetric $h[n]$ with M odd (the 6 mark answer)

- Start from the definition and pull out the centre term at $n = \alpha = (M-1)/2$:

$$H(e^{j\omega}) = \sum_{n=0}^{M-1} h[n] e^{-j\omega n}$$

- Substitute $n \rightarrow M-1-n$ in the sum and use $h[n] = h[M-1-n]$ to pair each term with its mirror image. Each pair contributes

$$h[n](e^{-j\omega n} + e^{-j\omega(M-1-n)}) = 2h[n] e^{-j\omega\alpha} \cos(\omega(\alpha - n))$$

- Summing the α pairs and the lone centre term factors the whole response:

$$H(e^{j\omega}) = e^{-j\omega\alpha} \underbrace{\left[h[\alpha] + 2 \sum_{n=0}^{\alpha-1} h[n] \cos(\omega(\alpha - n)) \right]}_{\text{real, call it } H_r(\omega)}$$

- The bracket is **purely real**, so the entire phase is the exponential in front:

$$|H(e^{j\omega})| = |H_r(\omega)|, \quad \arg H(e^{j\omega}) = -\alpha\omega \quad (\text{plus } \pi \text{ wherever } H_r < 0)$$

- **That is the proof**: the phase is a straight line of slope $-\alpha$, so the group delay is α samples at every frequency.

The antisymmetric case runs identically but the mirror pair subtracts, giving $2jh[n] \sin(\omega(\alpha - n))$ and hence $H = j e^{-j\omega\alpha} H_i(\omega)$. The extra j is the constant $\pi/2$ in the phase, which is why an antisymmetric filter has **generalized** linear phase rather than the plain kind. See §5 of the numerical companion for $h[n] = \{-1, 0, 1\}$, which is this case.

3.7 Last-minute recall

Say these without thinking

- $H(e^{j\omega}) = H(z)|_{z=e^{j\omega}}$, **provided the ROC contains the unit circle.**
- $|H| = \frac{\text{product of distances from the zeros}}{\text{product of distances from the poles}}$.
- Pole near the circle $\rightarrow$ peak; zero near the circle $\rightarrow$ dip; zero on the circle $\rightarrow$ exact null.
- Peak sits at the **angle of the pole**, dip at the **angle of the zero**.
- $|H(e^{j0})| = |B(1)/A(1)|$, $|H(e^{j\pi})| = |B(-1)/A(-1)|$.
- Causal and stable $\Leftrightarrow$ **all poles inside** $|z| = 1$.
- FIR: always stable, exactly linear phase possible, all poles at the origin.
- Linear phase $\Leftrightarrow h[n] = \pm h[M - 1 - n]$, delay $\alpha = (M - 1)/2$.

Mistakes that cost marks

- **Forgetting the zeros at the origin** when $M < N$. They belong on the plot even though they do not shape $|H|$.
- Drawing a magnitude response for a system whose ROC excludes the unit circle. If a pole lies outside and the system is stated causal, **the correct answer is that no frequency response exists.**
- Reading the poles off the x coefficients. **Poles come from the y side.**
- Losing the conjugate. A real difference equation can only have complex roots in **conjugate pairs**; a lone complex pole means an arithmetic slip.
- Sketching a peak at the pole's *radius* instead of its *angle*.
- Quoting ω in Hz. It is in radians per sample, and the plot runs 0 to π .

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. $\rightarrow$ reads "leads to / therefore", $+$ reads "and", $=$ reads "is". $\checkmark$ means the property holds, $\times$ that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months**: 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II).
Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 Bh is EX 753 Regular, 74 Ma is EX 753 Back.
- **69 Bh and 69 Bh are two different papers**, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.